

TPC-412: Pooled Complete-Shell Extension

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Abstract

We extend the pooled odd complete-shell proxy from one height to $H = 16, 32, 66, 128$, always with $N = 4H$. The complete shells at $Q = 65536$ and $Q = 131072$ retain all $5709+10749 = 16458$ primes, use shell-local amplitudes, and have equal pooled parity counts $m_- = m_+ = 8229$. Exact rational arithmetic and an independent literal replay verify all four rows. This is finite synthetic evidence, not an arithmetic or twin-prime theorem.

1 Profile and theorem

Let $\mathcal{P}$ be the increasing union of $(65536, 131072]$ and $(131072, 262144]$. For $p_i \in \mathcal{P}$, let Q_i be its shell scale and set $a_i = p_i^3/[Q_i^2(p_i - 1)]$. For each listed H , put $N = 4H$ and choose an origin above 10^6 with residues 0 on even indices and $-N$ on odd ones. All primes exceed the largest $N = 512$, so the even class masks offset zero, the odd class first hits offset N , and no prime masks offset one. Therefore

$$G_0 = V_- S_0, \quad G_1 = V_- S_1 + V_+(S_1 - t_1^2), \quad M = t_1 P_-.$$

Theorem. For every $H \in \{16, 32, 66, 128\}$,

$$0 \leq z = \frac{M}{\sqrt{G_0 G_1}} \leq \frac{t_1}{a_{\min} \sqrt{S_0 S_1}} \leq \frac{4}{a_{\min} H} \leq \frac{4}{H}. \quad (1)$$

Indeed, $P_-^2 \leq m_- V_-$, $V_- \geq m_- a_{\min}^2$, and $G_1 \geq V_- S_1$; also $S_0, S_1 \geq H/4$ and $a_i > 1$. This proves (1).

2 Exact observations

The exact pooled certificate reports

H	N	z
16	64	0.021785036050694
32	128	0.010809175951931
66	264	0.005216872683870
128	512	0.002683946067759

The independent checker rebuilds both sieves and the CRT, then literally visits every per-prime/per-coordinate mask for both local rows at every height.

3 Scope and reproduction

This finite four-height synthetic proxy does not prove a full normalized operator estimate, identify physical h_0 or arithmetic signs, pay arithmetic L^2 or fixed-power credit, close Route A/B, or imply a twin-prime result. Run the producer, independent replay, and mutation stress with `--check` in normal and optimized modes; Bridge-B repeats and locks these checks.